

Considering Fractions

Q1 *An Informative Fraction*

Whether we are interested in the opioid crisis, road safety, low birth weight babies, suicide, or vaccinations, we often work with fractions to measure the phenomenon. A fraction might be a proportion, like the proportion of babies born who weigh less than 250 grams; it might be a rate, such as the number of deaths per years of exposure to an environmental hazard; or a ratio of two fractions (or two rates) so we can compare, say, the vaccine uptake of two groups.

In this discussion, we examine fractions to understand how they are calculated, and we ask questions about where the numbers come from, how they are measured, who or what is being counted and isn't being included, and whether different numbers might be more informative.

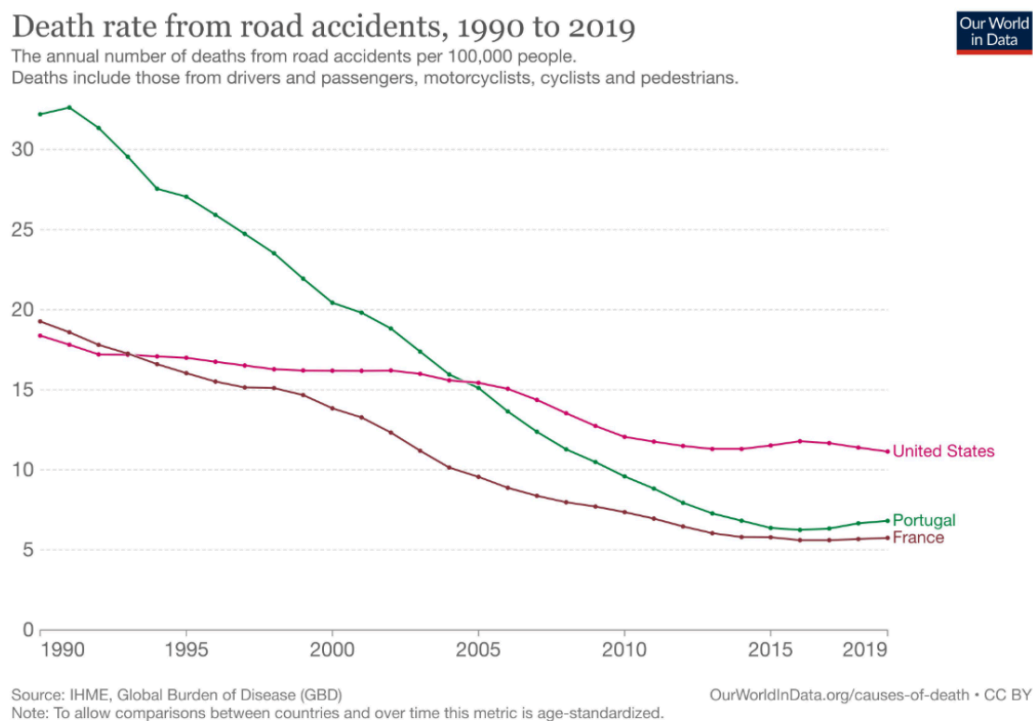


Figure 1: Figure 1: Screenshot from Krugman's tweet.

(Question 1 continued...)

Let's begin by looking at road safety. Nobel prize-winning economist Paul Krugman tweeted a chart (see Figure 1 and accompanying New York Times article) as he was winding up a long trip in Europe, including a visit to Portugal. The chart compared the road safety in Portugal to the U.S. and France over the past 50 years.

Let's focus on 2019 and the United States. Take a look at the scale on the y-axis to see what measurement is being plotted. It's a fraction:

$$\frac{\# \text{ deaths from road accidents in the US in 2019}}{\# \text{ of people (100,000s) in the U.S. in 2019}}$$

Q1.1 If there are roughly 323,800,000 people in the US in 2019, how many 100,000s of people are there?

Q1.2 Why doesn't the chart simply report the number of deaths in road accidents that year? For the U.S. that would be 36,096.

Q1.3 Why not simply use a fraction, like:

$$\frac{36096}{323800000} = 0.0001114762$$

Or a percentage such as:

$$100 * \frac{36096}{323800000} = 0.01114762\%$$

Q2 Digging Deeper: Where do numbers come from?

Q2.1 Answer these questions for the road safety example, and consider the numerator and denominator, separately. You may need to do some Internet research to get the answers.

- Who is doing the measuring?
- What is being measured (be as precise as possible)?
- Who is being measured? Who might be missed?
- What might be missed by this measurement?

Answering these questions helps us figure out whether the numbers are: trustworthy; measuring what we think they are measuring; and comprehensive.

Q2.2 The footnote for the figure mentions that “to allow comparisons between countries and over time, this metric is **age-standardized**.” Age standardization means that rates are calculated separately for each age group in a country and then combined according to a specific weighting, where the weighting is the same for each country. Can you think of a scenario where the comparison between countries would be misleading without age-standardization?

Q2.3 The tweets in response to Krugman raise several questions about who is missing and whether or not it is the right metric. Below is a sample of four responses to Krugman’s plot. Explain what each tweet is recommending: does it make a case for adjusting the ratio or using a different numerator or denominator?

1. “Seems like deaths per 100,000 Vehicle Miles Traveled, rather than per 100k population, might be a better comparison, given the need to drive everywhere in most of the US.”
2. “In the 70’s and 80’s there were almost no highways in Portugal. In the 90’s and 2000’s a lot were built, and fatalities decreased significantly.”
3. “DUI limit is 0.05 [in Portugal] vs 0.08 in US. Makes a huge difference. Sweden is 0.02 which is effectively do not drink at all if you’re driving.”
4. “But that is also a policy decision. Build safer forms of travel (rail, tram, segregated bike lanes) and fatalities will drop. Best metric would be deaths per km traveled across all modes not just cars.”

(Question 2 continued...)



Q3 Comparisons

We often want to disentangle a rate to see if the rate is roughly the same across different groups of people. These groups might be disaggregated according to age, income, race, sex, state, etc. In considering the road accident death rates, we might want to group the measurements by age to see if there's a case for raising the driving age or strictly enforcing tests for older drivers. Or, we might want to separate out the accidents that involved alcohol from those that did not. Below is a table from a report by the CDC that looks at fatal road accidents by age group.

This table breaks the population into 8 subgroups according to age and sex. The idea of looking at metrics for subgroups defined by two features (age and sex in this case) is called intersectionality.

Gender	Count 0-14	per 100k 0-14	Count 15-24	per 100k 15-24	Count 25-64	per 100k 25-64	Count 65+	per 100k 65+
Male	605	2.0	4149	19.0	16674	19.7	5243	21.8
Female	505	1.7	1718	8.2	5909	6.9	2790	9.3

Table 1: Fatal Road Accidents by Age and Sex (CDC)

Q3.1 Make three observations about these data.

Q3.2 Reflect on the tweeted critiques of the US - Portugal comparison and suggest an alternative. This may be a different rate, a feature to disentangle and compare the rates at a finer scale, or a group to include or exclude from the measurements. Justify your choice. What would you expect to see?

Q4 Energy Levels of Starship Pilots

Lisa, a starship fleet commander, is monitoring the energy levels of different pilots' starships. Each pilot starts with a certain energy level, represented by a number. Throughout the mission, the pilots' energy levels change based on various maneuvers and situations they encounter. However, Lisa's own starship energy level keeps changing as she mimics or adapts to the energy levels of other pilots. Lisa is now confused and needs help to figure out the final energy levels of each pilot.

(Question 4 continued...)

Below is the sequence of events. Determine the final energy levels for Lisa and her fellow pilots (Earn, Faith, and Richard) at the end:

1	richard = 100
2	faith = 120
3	earn = 150
4	lisa = 200
5	
6	lisa = 180
7	richard = lisa
8	
9	faith = 110
10	earn = faith
11	
12	lisa = 130
13	richard = faith
14	
15	faith = 90
16	lisa = faith
17	faith = lisa

Q4.1 Add comments to:

1	richard = lisa
2	earn = faith

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Q4.2 What are the final energy levels?

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Q5 Brewing up Some Code!

Elizabeth Fang, a well-renowned vampire, loves to brew magical potions in her free time. Her favorite potions are the Love Potion, which makes anyone who drinks it fall instantly in love with her, and the Invisibility Elixir, which grants her invisibility for 20 minutes at a time. Elizabeth is trying to calculate the total brewing time required for these potions over a week, assuming she needs to brew them daily.

Potion Type	Number of Potions Needed Daily	Average Brewing Time per Potion (minutes)
Love Potion	20	3.5
Invisibility Elixir	15	4.8

Table 2: Number of Potions and Average Brewing Time per Day per Potion

Elizabeth attempts to compute the **absolute value** of the difference between the total number of minutes spent brewing Love Potions and Invisibility Elixirs over a week (7 days). Unfortunately, she makes some errors in her initial code. Can you help her identify and correct these errors? Ensure that the corrected code outputs (shows the result) as a single, **positive** number.

HINT: You use the built-in function **abs** which takes in a number and returns the absolute value of that number.

```
1 data6 = 20(3.5)
2 brewing_time_invisibility = 15(4.8)
3 daily_difference = ((data6 - brewing_time_elixir)
4 weekly_difference = daily_difference * 7
5 total_time_difference = weekly_difference * 52
```

```
1 _____
   Q5.1
2 _____
   Q5.2
3 _____
   Q5.3
4 _____
   Q5.4
5 _____
   Q5.5
```